GAN-based Synthetic Radar Micro-Doppler Augmentations for Improved Human Activity Recognition

Baris Erol[1], Sevgi Z. Gurbuz[2], and Moeness G. Amin[1]
[1]Center for Advanced Communications, Villanova University, USA (berol, moeness.amin@villanova.edu)
[2]Dept. Electrical and Computer Engineering, University of Alabama, Tuscaloosa, USA (szgurbuz@ua.edu)

***Abstract*—Deep neural networks (DNNs), and, in particular, convolutional neural networks (CNNs), have recently received much attention in a wide range of research areas, including radar-based human activity recognition. However, obtaining a large training dataset still remains a challenging task due to the costs and resources required for data collections. In this paper, we present a method for extending adversarial learning to the generation of synthetic radar time-frequency (TF) domain signatures which provides the ability to adapt to different operational environments. The classification results achieved with a deep CNN trained on generated images prove the efficiency of proposed algorithm over the state of the art methods.**

***Index Terms*—human activity recognition, micro-Doppler, deep neural networks, generative adversarial networks**

I. Introduction

Radar for indoor monitoring is an emerging area of research and development, covering and supporting different health and wellbeing applications of smart homes, assisted living, and medical diagnosis. Radio frequency (RF) sensing provides viable information on human daily activities indoor as well as characterization of normal and abnormal gait [1]-[8]. The important properties of safety, reliability, low power, preservation of privacy, and insensitivity to light have expanded radar applications from open terrain target detection, classification, and localization to human motion recognition inside building and enclosed structures. The principal objective remains achieving high probability of detection and low false alarm rates.

Advances in human motion classification will have a critical impact on a broad range of applications. The population in general will benefit from better home safety and security, medical diagnosis, and a higher standard of living enabled by efficient human-machine interfaces, especially as it pertains to household appliances and devices. Irrespective of the sensing modality, progress in machine learning and Internet of Things (IoT) is rapidly changing what we expect from ubiquitous sensing. An RF sensor can act alone or in conjunction with others of the same modality, or/and within a suite of different sensors. Most significantly, RF sensing is non-invasive, non-contact, and effective even beyond obstacles, such as walls. When used for monitoring, radar can operate ambiently, in the background, without affecting a person's daily habits or routine.

Deep neural networks (DNNs) have shown great potential, and may well lead the way as a preferred motion classification choice in the near future [9]-[14]. However, a common challenge in radar-based human activity recognition is how to deal with the limitations of small datasets, especially when training DNNs, which are quite data-hungry for supervised learning [15]. Researchers have attempted to overcome this challenge by data augmentation, where the available radar data is modified through the operations of translations, time-shifting, and splitting signal records into small segments. However, these approaches may not necessarily lead to statistically independent training samples that effectively span the probable variations in target signature due to body size, speed, and individual gait characteristics. To overcome these limitations, a simulation methodology rooted in kinematic modeling through motion capture was proposed that instead applies data augmentation to the underlying skeletal model. This approach ensures generation of a large number of unique and kinematically consistent simulations spanning expected target profiles [16]. However, it does not provide a means for accounting for the variations in signal-to-noise ratio (SNR), clutter and other artifacts in the signature that may be caused by variable environmental conditions.

In this paper, we propose the use of a generative model, auxiliary classifier generative adversarial networks (ACGANs) [17] to generate large amounts of training data that can take into account environment-related effects, such as the clutter due to through-the-wall scattering. In this work, we show that synthesis of realistic human time-frequency signatures that include environmental artifacts using a generative model conditioned on external class labels yields an improvement of 14% in classification accuracy over that achieved by training with real data only.

The paper is organized as follows. In Section II, the radar motion database which is used to train the generative models, is introduced. In Section III, the generative model, ACGAN, is presented. In Section IV, the experimental results achieved by training a 19-layer CNN are presented in order to validate the generated databases. Finally, conclusions are provided in Section V.

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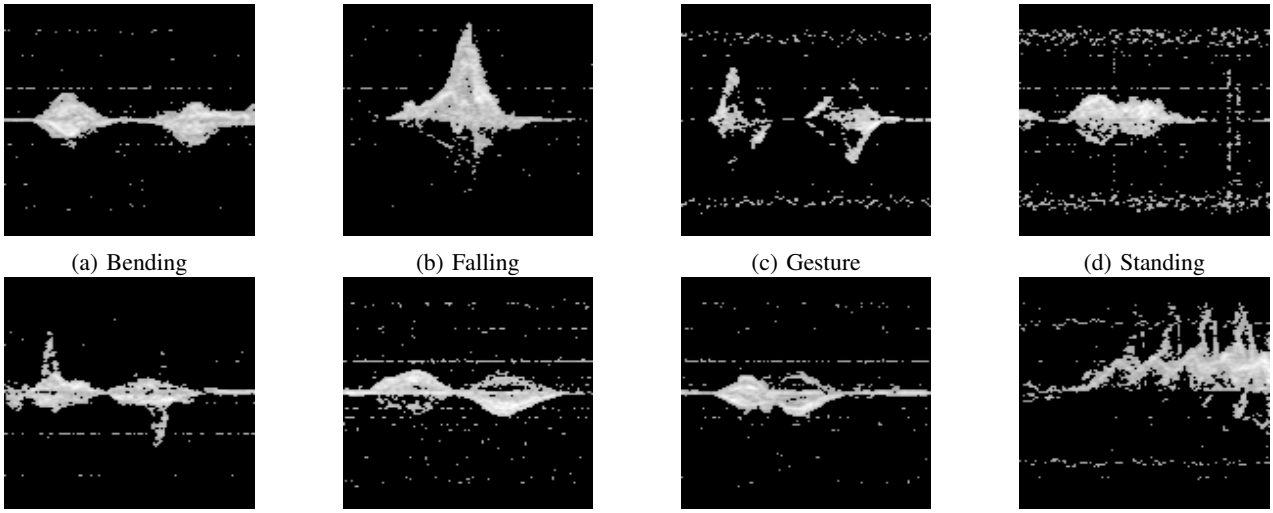

(a) Bending (b) Falling (c) Gesture (d) Standing

(e) Kneeling (f) Reaching (g) Sitting (h) Walking

Fig. 1: Real spectrogram images of different human activities (Horizontal and vertical axises depict time and Doppler frequency, respectively).

II. Micro-Doppler Database

A 25 GHz software-defined radar system developed by Ancortek, Inc. [18], SDRKIT 2500B, was used in this work to transmit a continuous-wave (CW) signal with 10 ms sweep time and 128 samples per sweep. As a result of the selected parameters, the received radar signal is highly oversampled at a rate of 12.8 kHz. The higher sampling frequency causes the spectrum to occupy a smaller bandwidth about 0 Hz, leaving much of the frequency range of the time-frequency transform devoid of any signal components. Therefore, as pre-processing, signals are first downsampled to 1.2 kHz.

The Doppler information of the human motion activities are often discriminative and descriptive of the nature of the signal, its source, and the mechanism of its generation. Time-frequency signal representation is the most commonly used analysis tool of nonstationary signals. Spectrograms expose the signal's local frequency behavior and are mathematically defined as

$$\mathrm{S}(n,k) = \left| \sum_{m=0}^{N-1} \mathrm{h}(m)\mathrm{x}(n-m)e^{-j2\pi km/N} \right|^2, \qquad (1)$$

where $\mathrm{h}(m)$ is a window function which can affect both the time and frequency resolutions. The spectrograms are also known as the micro-Doppler signatures, because they reveal the frequency modulations (micro-Doppler frequencies) caused by vibrations and rotations about the central Doppler shift due to translational motion [19]. In the case of humans, the intricate motion of limbs that occurs during the course of any activity, results in a visually recognizable micro-Doppler signature in the TF domain. The spectrograms of 8 different motions are provided in Figure 1-(a) through (h).

Two datasets were collected from two different indoor environments: Radar Imaging Lab (RIL) and Center for Advanced Communications (CAC) conference room, both places are located on Villanova University campus. The radar was placed on a table with a height of 3.2 ft for both of the locations. Real radar experiments were performed by 14 human subjects. A total of 1356 radar backscattering data were collected from the test subjects posed heights ranging from 5.1 to 6.3 ft, weights ranging from 119 to 220 lbs, and included 12 male and 2 female subjects. Eight different motions were considered in the experiments: bending (167), falling (120), kneeling (216), gesture (150), reaching (140), sitting (233), standing (130), walking (200). Motions were performed for three different directions, namely, 0°, 30°, 45°. Each subject performed the experiments with three different speeds: slow, medium, and fast. The slower motions are similar to what could be expected from the typical motion of elderly, who are a risk population that could most benefit from RF sensing.

III. Auxiliary Classifier Generative Adversarial Networks (ACGAN)

The term "generative" is used in many ways in machine learning community. Within the scope of this work, this term refers to a model that takes a training data with a distribution p_{data} and seeks to learn a close estimate of it, denoted as p_{model}. Generative models attempt to predict features given a certain external label, whereas, discriminative models try to predict a label of a given input data [20]-[22]. These models can be classified into two broad categories of explicit (VAE, PixelRNN/CNN [23], [24]) and implicit (GAN [25], Markov chain) approaches [26].

Generative models are typically used in improving the performance of reinforcement learning, domain adaptation, and overcoming the problems with missing data [26]. In this work, generative models are utilized to increase the amount of training data as well as to broaden the intra-class diversity. We consider a conditional generative model: ACGAN [17]. The model yielded by vanilla GAN can easily serve as a density model of the training data and it does not need any

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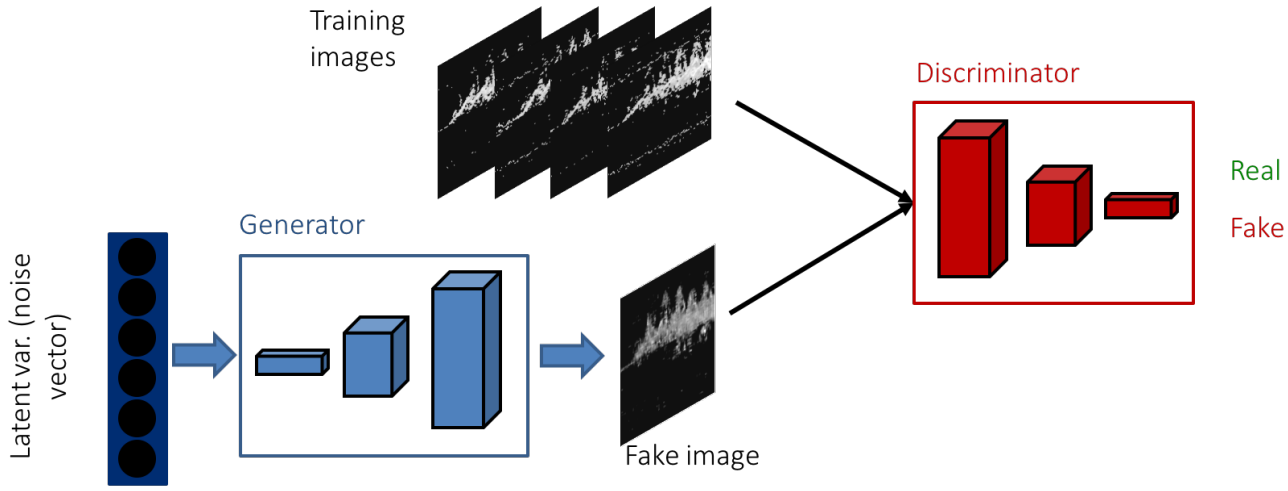

Fig. 2: GAN architecture.

class information. An example of the vanilla GAN structure is depicted in Figure 2. On the other hand, ACGAN allows the generative model to condition on external information (class labels). An alternative to this conditional model is to train N (number of classes) separate models. This has an adverse effect of bringing the problem of overfitting due to the small amount of available training data, and requires a high computational power.

ACGAN is an extension of the vanilla GAN model which enables the model to be conditioned on external labels to improve the quality of the generated images. One method to produce class conditional samples is to supply both generator and discriminator with class labels. However, the strategy behind the ACGAN is to, instead of feeding the class information to the discriminator, one can task the discriminator with reconstructing the label information. This builds on the fact that forcing a model to perform additional tasks or constraints improves performance of the original problem [17]. It is achieved by modifying the discriminator to contain an auxiliary decoder network that outputs the class labels for the training data. The objective function of the ACGAN has two parts: the log-likelihood of the correct source, L_{s}, and the log-likelihood of the correct class, L_{y}.

$$L_s = \mathrm{E}[\log p(\mathrm{s} = real|\mathrm{x}_{real})] + \mathrm{E}[\log p(\mathrm{s} = fake|\mathrm{x}_{fake})]. \quad (2)$$

$$L_{\mathrm{y}} = \mathrm{E}[\log p(\mathrm{Y} = y|\mathrm{x}_{real})] + \mathrm{E}[\log p(\mathrm{Y} = y|\mathrm{x}_{fake})], \quad (3)$$

where s is the generated images. The discriminator is trained to maximize the $L_s + L_Y$ while the generator is trained to maximize $L_Y - L_s$.

The employed ACGAN architecture consists of two different parts: the generator and the discriminator. The generator takes a vector of 100×1 random numbers drawn from a uniform distribution and class labels as inputs and outputs a spectrogram image of size $64 \times 64 \times 1$. We used a similar generator network as in [17] with minor modifications for generating human activity spectrograms. The generator consists of a fully connected layer reshaped to size $4 \times 4 \times 128$ and four 2D convolutional layers with 3×3 kernel size and 128 filters each. Batch normalization and 2D up-sampling (stride of 2×2) are applied to each layer of the generator network, except for the output layer. Addition to the batch normalization, dropout of 0.15 is also applied in each layer considering the small amount of real training data. Adding this regularization into the generator network helps combat overfitting and mode collapsing. ReLU activation functions are applied to all convolutional layers except the output layer which employs a *tanh* activation function. The discriminator structure consists of layers similar to a VGG16 [27] network with the addition of dropout of 0.15 (every even layer) and utilization of LeakyReLU as the activation function in each layer except for the last. The last layer of the discriminator uses a sigmoid for the validity of the generated images and softmax for reconstruction of the class labels.

Pre-processing includes implementation of a cleaning and filtering algorithm, followed by scaling the images between $(-1, 1)$ for *tanh* activation function. The weights are initialized with a normal distribution. The ADAM optimizer, an adaptive moment estimation method, is utilized with a learning rate of 0.0002, $\beta_1 = 0.5$, and $\beta_2 = 0.999$ for 3000 epochs and minibatch size of 16. Some examples generated by the proposed ACGAN are depicted in Figure 3. These synthetic samples exhibit similar kinematic patterns as that normally observed in real training samples, as shown in Figure 1. More interestingly, ACGAN successfully learns how to generate spectrograms for the cases that are not on the real training database. One specific example is for the falling class, where none of the real training experiments were performed away from the radar (resulting in negative frequency components). However, as it can be clearly seen in Figure 3, there are some

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synthetic samples which correspond to the fall motion being performed away from the radar. A total of 40000 spectrogram images are generated using ACGAN (5000 for each class).

IV. Experimental Results

In this section, we present and compare the test classification performances achieved by a 10-layer DCNN trained on real spectrograms (DCNN), and DCNN trained on the database generated by ACGAN (ACGAN-DCNN). In order to analyze the improvement achieved by generative model, we collected a challenging test set in a completely different environment and configuration from the real samples (mentioned in Section II) that were used to train the ACGAN. The test set was collected from a male subject, who wasn't part of the training set, in a through the wall (TWR) setup, for multiple aspect angles (0°, 30°, 45°, 60°) to acquire a total of 385 samples.

The average test accuracies for DCNN and ACGAN-DCNN were found to be 79.3% and 82.56%, respectively. Thus, the ACGAN-DCNN performance is significantly higher than that achieved with existing, commonly used algorithms, and shows the advantage of using adversarial learning for training data generations.

The confusion matrix resulting from the ACGAN-DCNN approach is provided in Table I. The primary source of confusion is between bending and kneeling as expected. These two motions have the same kinematic structure in the TF domain with one significant difference: motion of the knee. However, in some test experiments, motion of the knee was not that pronounced, resulting in this confusion.

V. Conclusion

In this work, we considered a generative model, referred to as auxiliary classifier generative adversarial network (ACGAN), for training of deep neural networks (DNNs). These networks are often limited by a lack of availability of sufficient RF data to support construction of deep networks beyond just a few layers. The classification performances achieved by training on real data versus training on synthetic data generated by an ACGAN are compared. It was shown that use of ACGAN-generated synthetic motion signatures resulted in a %14 improvement in classification accuracy.

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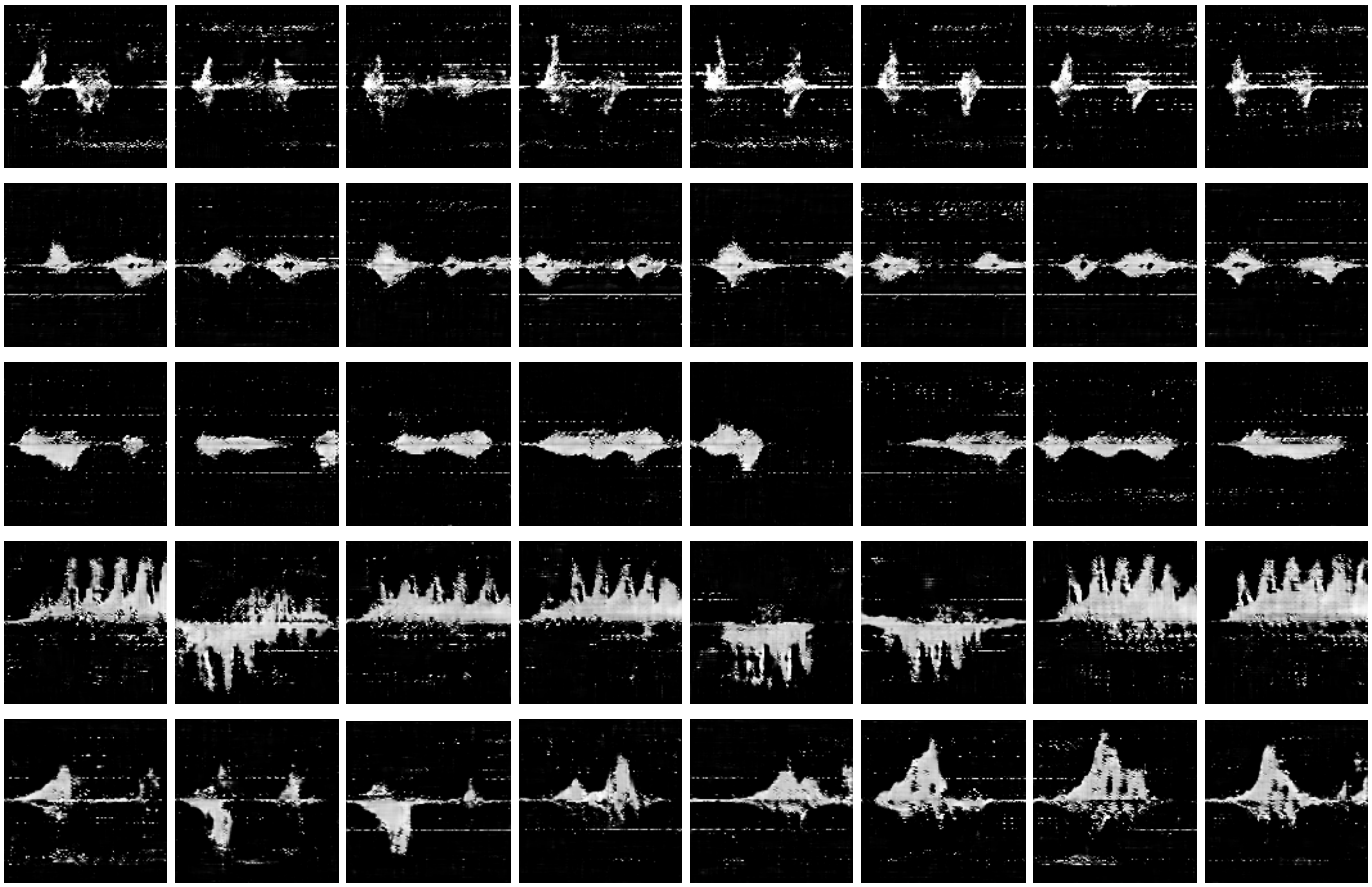

Fig. 3: Randomly chosen 8 samples generated by the ACGAN. Each row represents a different class (gesture, bending, sitting, walking, falling).

TABLE I: Test confusion matrix for ACGAN-DCNN with overall test accuracy of 82.56%.

%	Bending	Falling	Gesture	Kneeling	Reaching	Sitting	Standing	Walking
Bending	**90**	0	0	0	0	8	0	2
Falling	2.7	**81**	2.7	1.3	1.3	2.7	0	8.3
Gesture	0	0	**86**	8	0	0	0	6
Kneeling	30	0	0	**70**	0	0	0	0
Reaching	6	0	0	0	**74**	16	0	4
Sitting	0	0	0	0	0	**100**	0	0
Standing	0	0	2	4	2	8	**66**	18
Walking	0	0	0	6	0	0	0	**94**

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